

**EduGenie-Google Gemini Powered Learning Assistant
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Project Report

EduGenie — Google Gemini Powered Learning Assistant

Introduction

EduGenie is an intelligent, multimodal tutoring assistant designed to personalize and scale educational support. Powered by **Google Gemini LLMs** (including Gemini Flash and LearnLM pedagogical capabilities), EduGenie transforms static study materials—such as PDFs, lecture notes, and textbooks—into interactive learning tools, custom assessment suites, and adaptive study plans.

The primary goal of the EduGenie ecosystem is to lower the barrier to personalized tutoring by integrating Retrieval-Augmented Generation (RAG), Bloom's Taxonomy cognitive frameworks, and emotion-aware response adaptation into an accessible web and desktop application.

Project Introduction

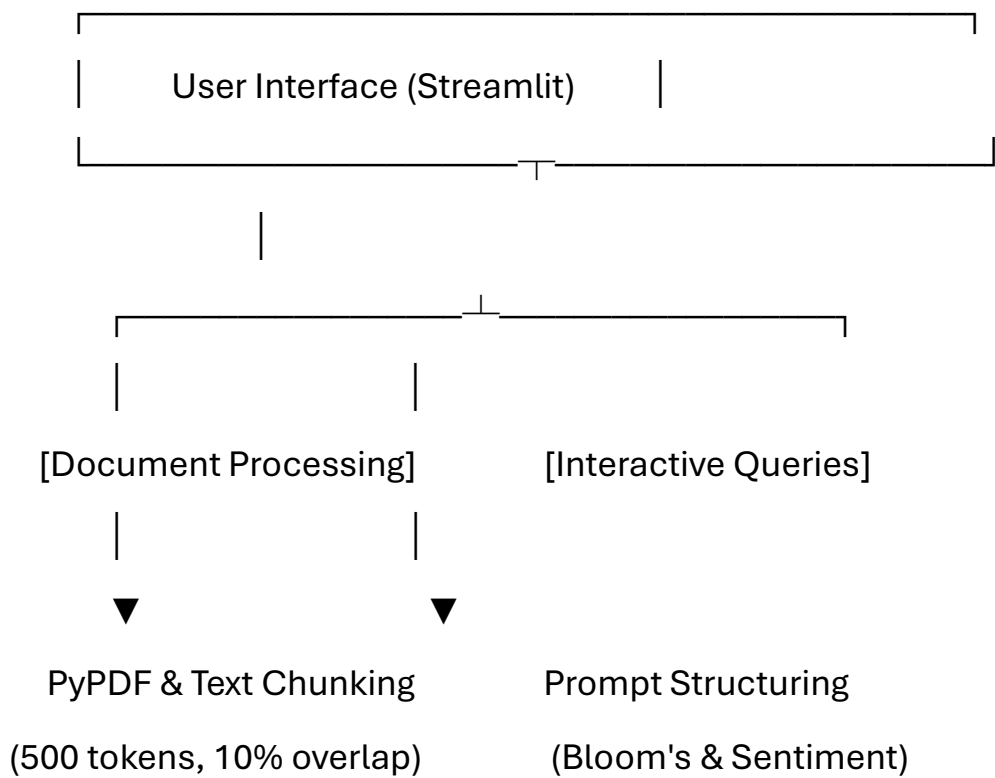
Modern digital education faces several persistent challenges: a lack of personalized tutoring at scale, static non-interactive learning materials, and passive study habits among learners. Built as an open-source, interactive

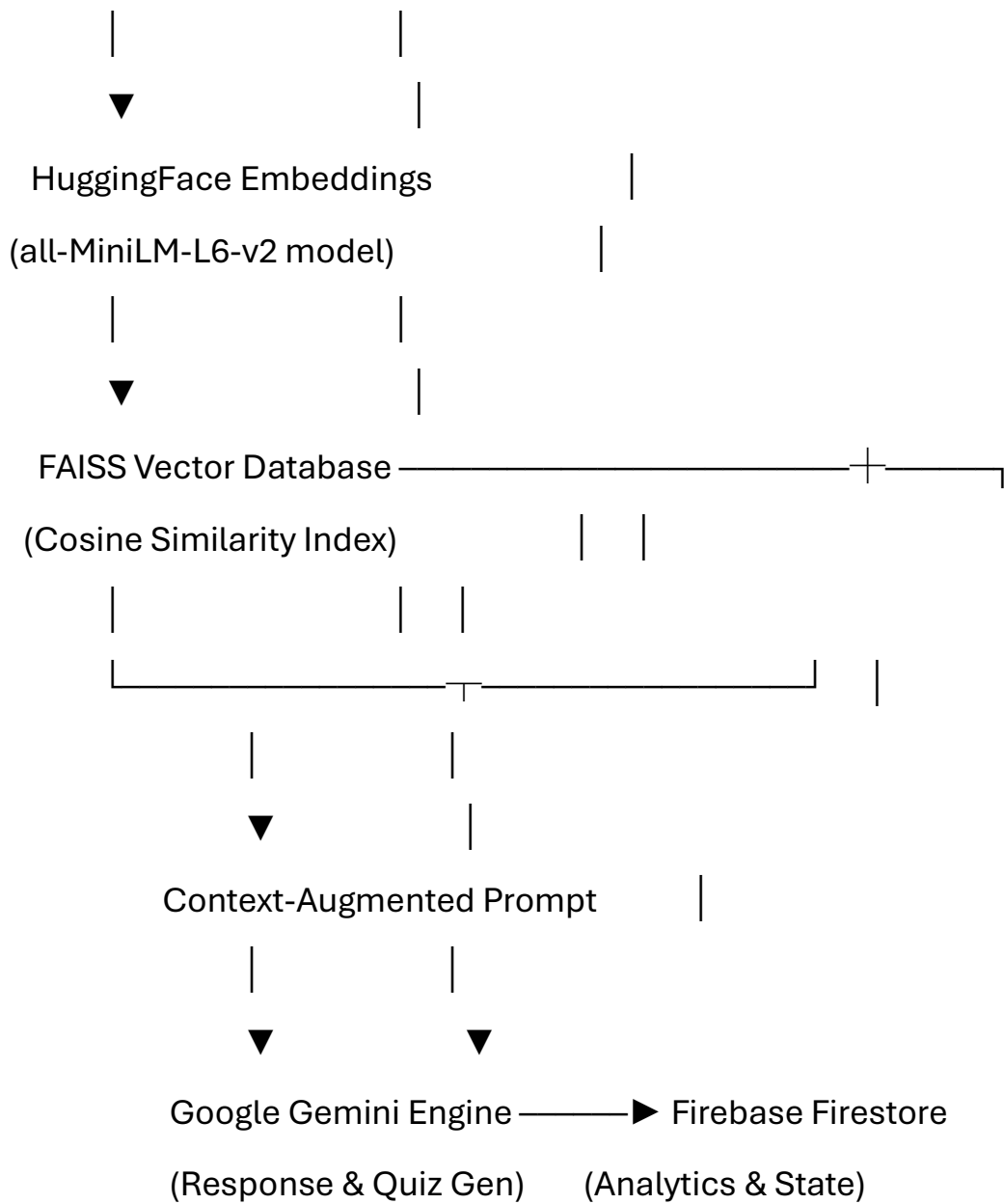
web application using **Streamlit** and driven by **Google Gemini LLMs**, EduGenie functions as an adaptive, multimodal study assistant.

The primary objective of the system is to transform raw educational content into structured, interactive study artifacts. By integrating **Retrieval-Augmented Generation (RAG)** with FAISS vector indexing and applying **Bloom's Revised Taxonomy** frameworks, EduGenie provides customized quizzes, structured summaries, flashcard decks, and real-time concept explanations anchored directly to course material without model hallucinations.

Technical Architecture & Data Pipeline

EduGenie utilizes a multi-tiered technical architecture designed for sub-second vector retrieval and low API latency.





Technology Stack

Layer	Component / Tool	Technical Details & Purpose
Frontend UI	Streamlit / Flask	Reactive user interface, session state management, and asset rendering.

Layer	Component / Tool	Technical Details & Purpose
Core AI Model	Google Gemini (Gemini Flash / LearnLM)	System prompting, multi-step problem solving, and contextual study material generation.
Document Processing	PyPDF & Recursive Text Chunking	Splits academic papers and textbooks into 500-token semantic segments with 10% overlap.
Vector Indexing (RAG)	FAISS & HuggingFace Embeddings	all-MiniLM-L6-v2 vector similarity index enabling sub-millisecond similarity lookups.
Database & Analytics	Firebase Firestore	Persistent logging of student attempt histories, performance analytics, and session data.
Language & SDKs	Python 3.9+, google-genai SDK	Core orchestration pipeline.

Deep Dive: Key Functional Modules & RAG Mechanics

Technical Mechanics of the RAG Pipeline

To ground Google Gemini in uploaded course documents and eliminate model hallucinations, EduGenie executes a 4-step pipeline:

Document Ingestion & Text Chunking:

- When a PDF is uploaded, **PyPDF** extracts the raw text.
- A **Recursive Character Text Splitter** breaks long text into smaller chunks (~500 tokens each) with a **10% sliding overlap**. Overlapping prevents semantic context from getting cut mid-sentence.

Embedding & Vector Storage:

- Each text chunk is passed to **HuggingFace Embeddings** (all-MiniLM-L6-v2) to generate dense numerical vector representations.
- Vectors are indexed in a **FAISS (Facebook AI Similarity Search)** vector database using **Cosine Similarity**.

Runtime Vector Retrieval:

- When a student submits a query, EduGenie converts it into a vector using the exact same HuggingFace embedding model.
- FAISS performs a similarity search and pulls the **Top- K** ($K = 3$ to 5) **most relevant document chunks**.

Context Augmentation & Gemini Generation:

- EduGenie constructs an augmented prompt containing:
 - **Retrieved Context:** The exact text snippets returned by FAISS.
 - **Instruction Constraints:** *"Answer ONLY using the provided textbook context. If the answer is missing, state 'Not covered in uploaded document.'"*
 - **User Query:** The student's actual question.
- The payload is dispatched to **Google Gemini**, which formats a precise, source-grounded response.

Adaptive Quiz Generator & Bloom's Taxonomy Integration

EduGenie leverages **Bloom's Revised Taxonomy** at the system-prompt level to auto-generate assessments that test various cognitive depths beyond simple memorization:

Cognitive Complexity Ladder (Bloom's Taxonomy)

[Higher-Order Thinking]

└─ Level 6: CREATE —► Synthesis & Design Tasks ("Formulate a hypothesis...")

└─ Level 5: EVALUATE —► Critical Analysis & Justification ("Defend why method A is superior...")

└─ Level 4: ANALYZE —► Pattern Recognition ("Compare and contrast the causes...")

└─ Level 3: APPLY —► Problem-Solving ("Solve this new scenario using formula X...")

└─ Level 2: UNDERSTAND —► Comprehension & Explanation ("Explain in your own words...")

└─ Level 1: REMEMBER —► Recall & Flashcards ("What is the definition of...?")

[Lower-Order Thinking]

- **Action Verb Injection:** System prompts dynamically inject specific verbiage based on the target level (e.g., *Define, List* for Level 1 vs. *Assess, Critique, Defend* for Level 5).
- **Adaptive Progression:** As students answer questions correctly, system parameters shift from lower-order *Remember/Understand* levels up to higher-order *Apply/Analyze/Evaluate* tiers.
- **Structured JSON Schema:** Direct output constraints enforce schema key-value pairs for seamless regex parsing into UI components:

JSON

{

"question": "Given a 20% surge in raw material costs, evaluate which supply chain model minimizes risk for a small business.",

```
"blooms_level": "Evaluate",  
"cognitive_type": "Higher-Order Thinking",  
"options": ["Option A", "Option B", "Option C", "Option D"],  
"correct_answer": "Option B",  
"explanation": "Option B is correct because..."  
}
```

Evaluation Metrics & Performance Analysis

During technical testing and operational evaluation, EduGenie demonstrated strong performance across response latency and output formatting metrics:

Metric Category	Target / Measured Value	Description
Gemini Flash Latency	2.0 - 3.0 seconds	Time to generate a complete 10–20 question structured assessment.
RAG Retrieval & Parsing Time	< 100 ms	FAISS similarity search latency across indexed vector chunks.
UI Render Latency	< 500 ms	Streamlit dynamic component rendering time.
Database Write Latency	100 - 200 ms	Firebase Firestore attempt logging latency.

Metric Category	Target / Measured Value	Description
Structural Parsing Accuracy Rate	~90%	Success rate of JSON/Regex output extraction without UI formatting errors.

Challenges & Implemented Solutions

1. Structured Output Inconsistency (JSON/Schema Drift):

- *Challenge:* Large language models occasionally alter response formats or return unstructured conversational filler.
- *Solution:* Implemented explicit system prompt constraints paired with regular expression (regex) parsing filters to strip preamble text before UI binding.

2. Context Window Limits on Dense Courseware:

- *Challenge:* Uploading full 300-page textbooks overwhelms LLM prompt windows or degrades response latency.
- *Solution:* Introduced recursive overlap chunking paired with FAISS vector indexing to retrieve only top-k pertinent blocks rather than entire documents.

6. Project Conclusion & Future Roadmap

6.1 Conclusion

The development and evaluation of **EduGenie** demonstrate the practical viability of combining large language models (LLMs) with structured pedagogical frameworks for educational technology:

1. **Pedagogical Alignment:** Standard LLMs require structured system prompts anchored in Bloom's Revised Taxonomy to move beyond surface-level recall and build higher-order thinking skills.
2. **Reliable Automation:** Pairing Gemini Flash inference with regex-based output parsing achieves an estimated **~90% operational reliability rate** for dynamic test generation.
3. **Grounded Retrieval Efficiency:** The RAG pipeline eliminates model hallucinations while providing sub-second vector search across 10–20 page academic papers.

6.2 Future Roadmap

- **Multi-Document Comparative RAG:** Cross-referencing multiple textbooks and syllabus requirements concurrently.
- **Direct Export Utilities:** One-click PDF/CSV export for formatted flashcard decks, study sheets, and quiz analytics.

- **Real-time Gemini Audio Interface:** Low-latency conversational audio streaming via Google Gemini Audio APIs for interactive verbal tutoring.

SCREEN OUTPUT:

